

SHELLFORGE

Review 2 | Safe natural-language Bash command proposals

CLOUD CPU EXPERIMENT / 09 OCT 2026 / NO COMMAND EXECUTION

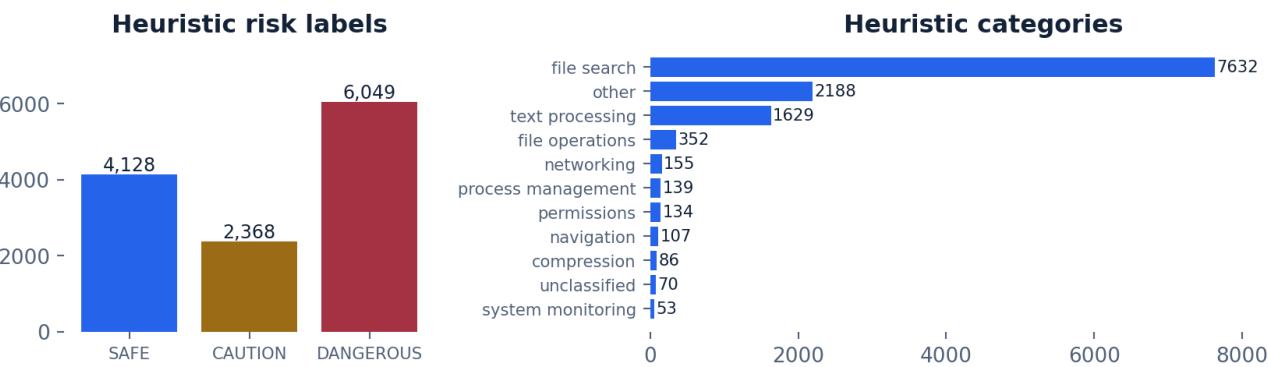
1. Dataset

The original NL2Bash corpus has 12,607 natural-language lines and 12,607 command lines. The repository also has 20 curated records, a 100-row import, 120-row pilot and old 96/12/12 splits. All original data and Review 1 documentation were preserved. Review 2 ingested 12,967 occurrences across the corpus and original JSONL files. Count equality and source SHA-256 hashes establish positional consistency, not universal semantic alignment. Original risk annotations remain unverified; independently verified corpus risk labels: 0.

Aligned corpus	Retained records	Train / Val / Test
12,607 pairs	12,545	10,036 / 1,255 / 1,254

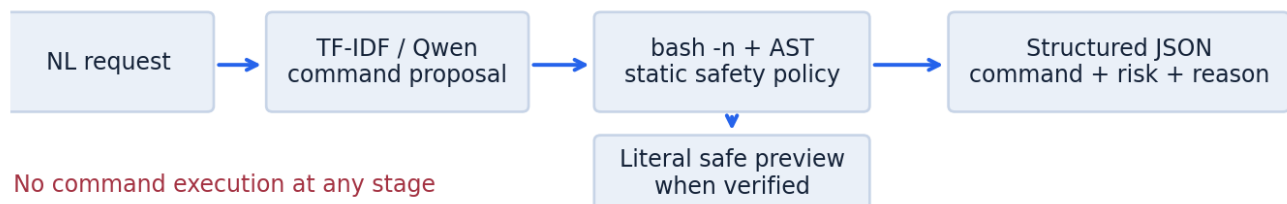
2. Dataset Preprocessing

Typed JSON validation rejects malformed objects, missing/non-string/empty required fields and NULs. Command bytes, case, quotes and internal spacing are preserved. Syntax checks use Bash parse-only mode with a clean environment. Removed 348 duplicate occurrences and 74 syntax-invalid occurrences; retained 12,545. Connected components isolate shared normalized instructions, AST literal quote/spacing equivalents and known additive ls/rm flag variants. Seed 42 yields 9,328 groups (largest 45); splits 10,036/1,255/1,254. All checked cross-split overlaps are zero. Arbitrary semantic paraphrase leakage is not ruled out. Missing optional fields before dedup: category 12,607, risk 12,607, explanation 12,907, safe alternative 12,949; retained as missing/provenance rather than fabricated annotations.



All distribution labels shown above are heuristic. Originals and old splits remain intact; full lineage and discard reasons are saved separately.

3. Code Implementation



A training-only character TF-IDF retrieval baseline and optional Qwen Transformers generator feed the same static validator and structured JSON CLI. The validator handles deletion flag variants, find actions, file truncation, formatting, privilege changes, broad permissions, termination, wrappers, interpreters, substitutions and dangerous pipelines. Unknown utilities require review; unsupported shell constructs fail closed. A preview is emitted only for simple literal recursive rm and is revalidated as read-only. Other cases explicitly return no verified preview. Every response records `executed=false` and requires manual review. Completed a small CPU LoRA pilot: 20 optimizer steps, rank 8, alpha 16, q_proj/v_proj; 540,672 trainable parameters (0.1093%). Used 128 of 128 selected training examples; 0 exceeded the 192-token limit. Validation loss: 1.3759. This is a feasibility run, not a fully trained production model.

src/preprocess_dataset.py | rendered source excerpt (not an app screenshot)

```
49 def import_pairs(nl, cm):
50     a, b = Path(nl).read_text().splitlines(), Path(cm).read_text().splitlines()
51     if len(a) != len(b):
52         raise ValueError(f'Positional alignment mismatch: {len(a)} NL vs {len(b)} command
lines')
53     return [{('instruction':i,'command':c), {'path':str(nl)+' ' + str(cm),'line':n}}
54             for n,(i,c) in enumerate(zip(a,b,1))], len(a)
55
```

src/inference.py | rendered source excerpt (not an app screenshot)

```
54 def propose(request,generator):
55     if not isinstance(request,str) or not request.strip():
56         raise ValueError('Request must be a nonempty string')
57     if len(request)>4000:
58         raise ValueError('Request exceeds 4000 characters')
59     proposal=generator.generate(request)
60     safety=check_command(proposal['command'])
61     preview=safe_preview(proposal['command']) if safety['risk']!='SAFE' else None
62     return {'request':request,**proposal,'safety':safety,
63           'explanation':proposal['explanation']+' '+'Policy findings: '+
'.join(safety['reasons']) if safety['reasons'] else 'Only allowlisted read-only utilities were
recognized.'},
64           'safe_preview':preview,'preview_explanation':'Lists literal removal targets; does
not simulate recursive contents.' if preview else 'No verified automatic safe preview
available.',
65           'executed':False,'requires_manual_review':True,
```

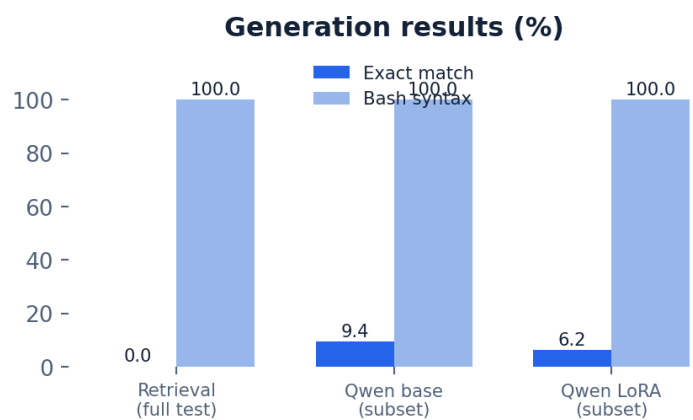
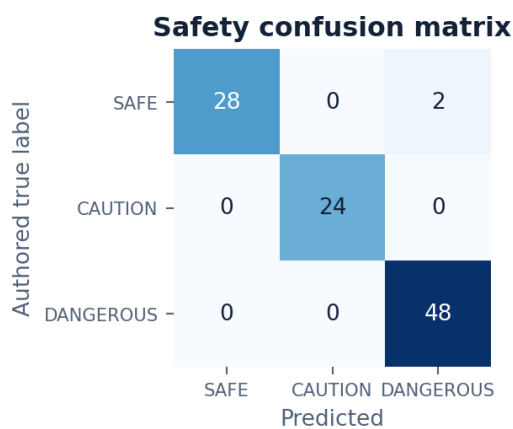
Evidence: real excerpts rendered from committed source, not application screenshots. CLI emits command, safety reasons, syntax validity, explanation, optional preview, backend and `executed=false`.

4. Metrics - Results and Discussion

Generation is evaluated on held-out requests using strict command-byte exact match and Bash -n syntax validity. The retrieval baseline has 0.00% exact match on 1254 cases; command grouping intentionally prevents memorized command overlap. Its 100.00% syntax validity reflects copying valid training commands, not task correctness. Qwen base/LoRA comparisons, when present, use the same deterministic 32-record held-out subset, not the full test. Safety uses 102 separately authored cases with operation-based labels, never validator-generated truth. Final macro F1 is 0.9817; dangerous false negatives: 0. This benchmark was used to refine the policy and is not an untouched generalization set. Labels are assistant-authored, not external expert consensus. No command execution or semantic correctness percentage is claimed. On the same 32 requests, Qwen base exact match was 9.38%, versus 6.25% for LoRA. The small fine-tuning run did not improve exact match; this subset is too small for a robust generalization claim.

Model / evaluation scope		n	Exact match	Bash syntax
TF-IDF 1-NN / full test		1254	0.00%	100.00%
Qwen base / subset		32	9.38%	100.00%
Qwen LoRA / same subset		32	6.25%	100.00%

Safety class	Precision	Recall	F1	Support
SAFE	1.000	0.933	0.966	30
CAUTION	1.000	1.000	1.000	24
DANGEROUS	0.960	1.000	0.980	48



Exact-match success: Make directory "temp"

Reference: mkdir temp | Proposed: mkdir temp

Exact-match failure: Find all files starting from the current directory which are exactly 100MB in size

Reference: find . -size 100M | Proposed: find . -size +100M

5. What Went Wrong

The original importer could silently truncate unequal files; old random splits leaked shared commands/instructions. Retrieval produced zero held-out exact matches, showing its inability to synthesize unseen commands. The initial safety benchmark missed `date -s` and `chmod ugo=rwx` (two dangerous false negatives); both were fixed while preserving initial scores. Two harmless compound examples remain rejected by the conservative policy. Direct git clone failed authentication; connector access recovered the repository. Initial model-client dependencies were incompatible with the available proxy; pinned compatible versions resolved access. An initial training run was interrupted to finalize stronger split equivalence, then restarted on final artifacts. The 20-step CPU pilot uses very little data and cannot establish robust model quality. LoRA exact match decreased from 3/32 to 2/32; a completed training run is not proof of improvement.

6. Alternative Flow / Proposed Improvements

Future work: externally adjudicate corpus pairs and safety labels; reserve a fresh safety holdout after policy development; expand AST analysis with argument-aware utility contracts and paraphrase clustering; test adversarial/obfuscated syntax. Run larger GPU LoRA/QLoRA experiments with identical split hashes, token budgets and a shared untouched evaluation set; compare base/tuned accuracy with uncertainty intervals. Add human semantic review of generated commands and calibrated abstention. Keep execution disabled: the system should remain a reviewed command-proposal tool. Completion evidence is stored in JSON metrics, example JSONL, source hashes, actual logs, training manifest and adapter files; no fake UI screenshots are used.

Verification and reproducibility

===== 37 passed in 16.56s
=====

Cloud runtime: Linux x86_64, 9 CPUs, no CUDA. Model revision: 7ae557604adf67be50417f59c2c2f167def9a775. See README.md for dependency installation and exact commands; use scripts/reproduce.sh for baseline, tests and report regeneration.

Evidence	Repository path
Source/data audit and counts	docs/review2_audit.md; data/review2/stats.json
Actual generation + safety predictions	reports/metrics/; reports/logs/
LoRA completion and model artifact	artifacts/qwen-lora/training_manifest.json
GPU continuation (future larger run)	notebooks/ShellForge_Colab.ipynb

References: Lin et al., NL2Bash (2018), github.com/TellinaTool/nl2bash; Hu et al., LoRA (2021), arxiv.org/abs/2106.09685; Qwen2.5 model card, huggingface.co/Qwen/Qwen2.5-0.5B-Instruct. Corpus file alignment is positional evidence only; no renewed claim of independently verified labels.